Explaining and Predicting the Operating Efficiency

of United States Banks Using FDIC Financial Data

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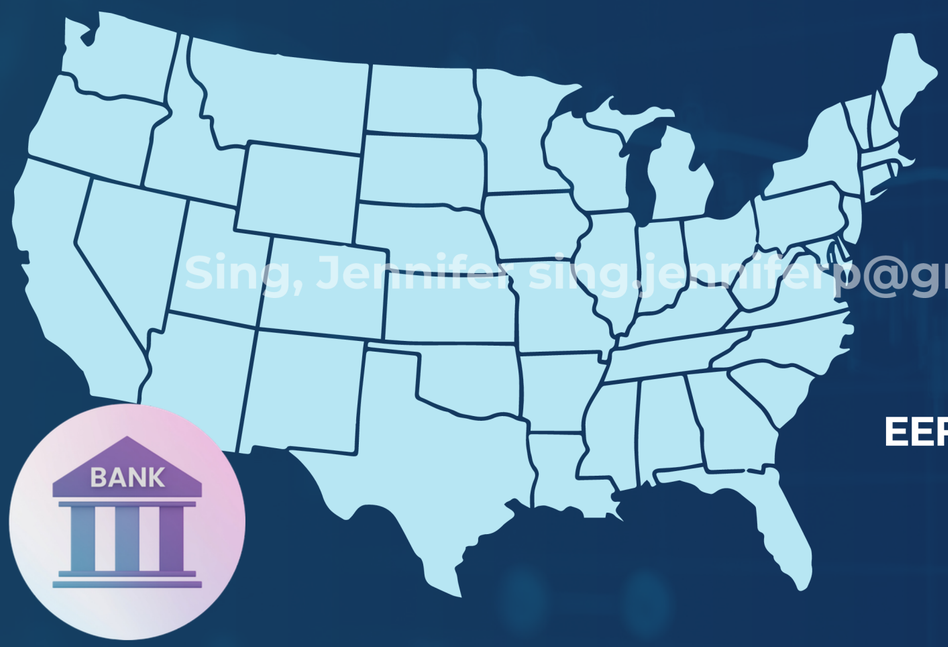
U.S. Bank Operating Efficiency



EEFFR

Efficiency Ratio

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$$\text{EEFFR} = \frac{\text{Non-Interest Expense (Operating costs)}}{\text{Total Revenue (Net interest income + non-interest income)}} \times 100\%$$

Lower value means a leaner operation.

Internal & Management Driven; Unlike Credit losses

Key metric that senior managers watch closely.

EEFFR = FDIC variable name; ER = Efficiency Ratio. Both refer to the same measure.

A lower efficiency ratio signals a leaner bank with stronger profit potential.

Beyond the Ratio: From Challenges to Strategy

From identifying controllable drivers to forecasting efficiency, our analytics support targeted management decisions.

What drives a bank's operating efficiency?

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Quantifying drivers lets leadership move from blanket cuts to targeted interventions

Can next-year Q1 efficiency be predicted using current-year Q1 financial data?

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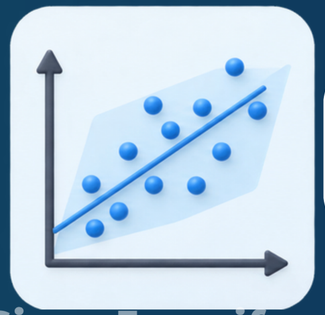
Forecasts provide early-warning signals for regulators and investors

Challenge: A single efficiency ratio is hard to interpret without knowing bank size & business model context

Predictive Modelling for Driver Analysis & Forecasting

Combining interpretable linear models and ensemble trees to quantify efficiency drivers and predict performance.



Multiple Linear Regression

Random Forest

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Decision Tree

XGBoost

Validation Strategy: Expanding Window Cross Validation and Hyperparameter Tuning

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End-to-End Data Warehousing Pipeline

BUSINESS OBJECTIVES

- Monitor operating costs through interactive dashboards
- Track changes in the Efficiency Ratio (ER) over time
- Identify cost trends and key factors affecting efficiency
- Support data-driven decision-making

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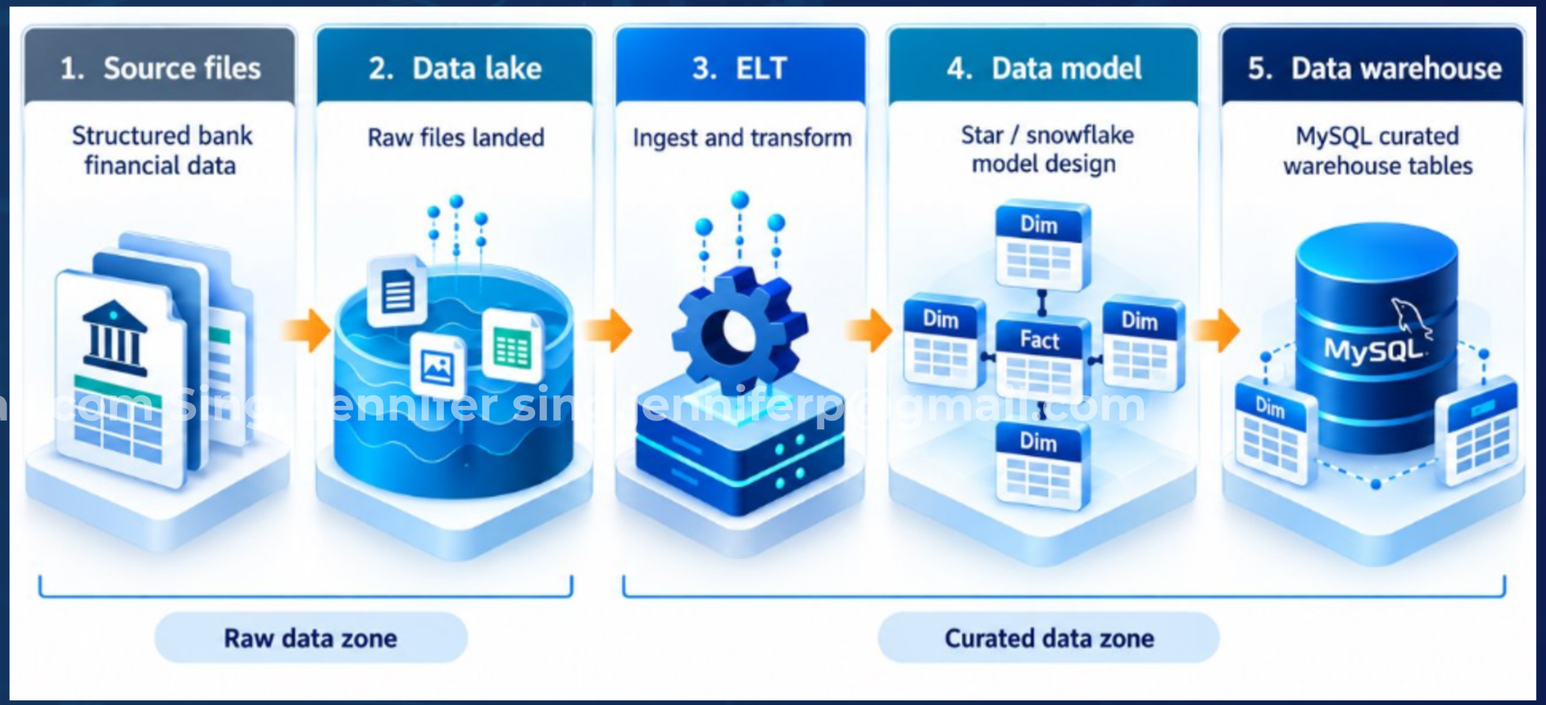
DATA ARCHITECTURE

Implement a data lake and data warehouse architecture to support scalable data storage, processing, and reporting.

DATA MODEL

Use a star schema to provide fast, consistent, and business-friendly reporting.



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From Medallion Architecture to Star Schema

Raw data → Validated history → Analytics-ready warehouse

FDIC Call Reports

BRONZE

Raw Delivery Staging
- Replace the previous delivery
- Retain all source columns
- Validate bank-quarter keys
- Tag each row with batch ID

SQL-based Data Processing

SILVER

Validated Delivery History
- Keep all source columns
- Preserve earlier versions
- New token for corrections
- Same-batch retry: no inserts

GOLD

Fact & dimensions
Star schema
Reporting ready

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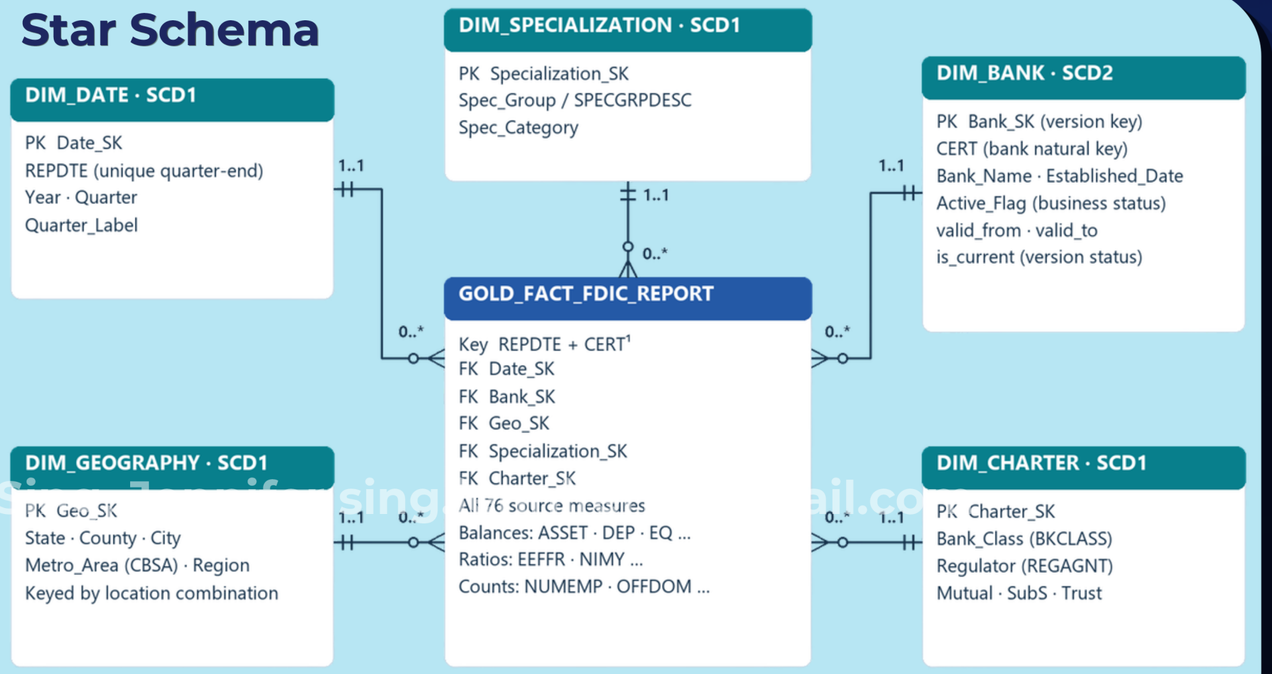
Star Schema



Grain: one bank (CERT) per reporting quarter (REPDTE)

Data Warehouse

MySQL

BI & Reporting
Bank efficiency
Quarterly comparisons

Analytics & ML
Curated bank-quarter
financial measures

From FDIC API to ML-Ready Dataset

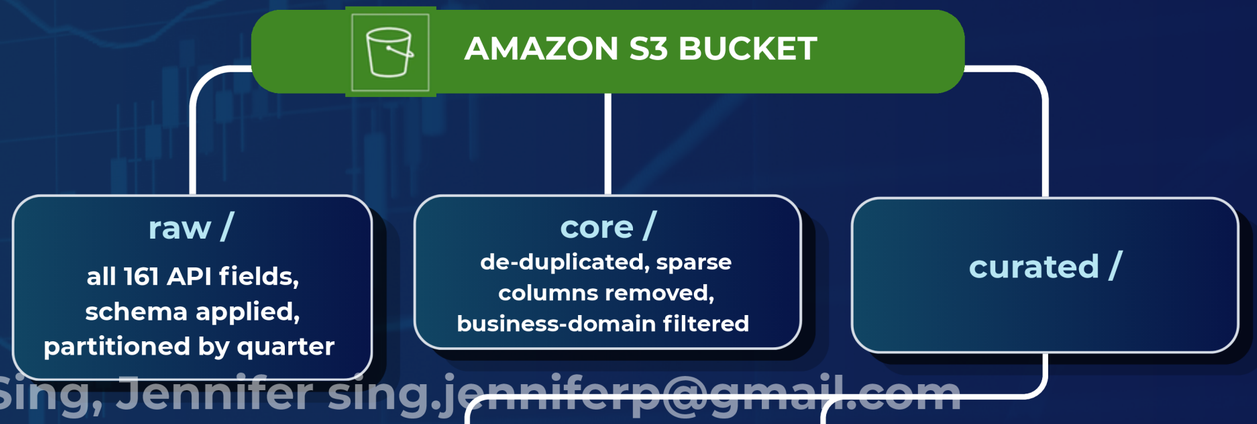
18 quarters of FDIC Call Reports prepared for predictive modeling.

FDIC BankFind Suite API api.fdic.gov/banks/financials

Q1 2022 – Q2 2026 = 18 quarters

Quarterly Call Report financials

83,041 total bank-quarter rows × 161 API fields



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Data Quality

- 88-92% completeness
- 0 duplicates
- Key identifiers complete

Feature Profile

73 predictors

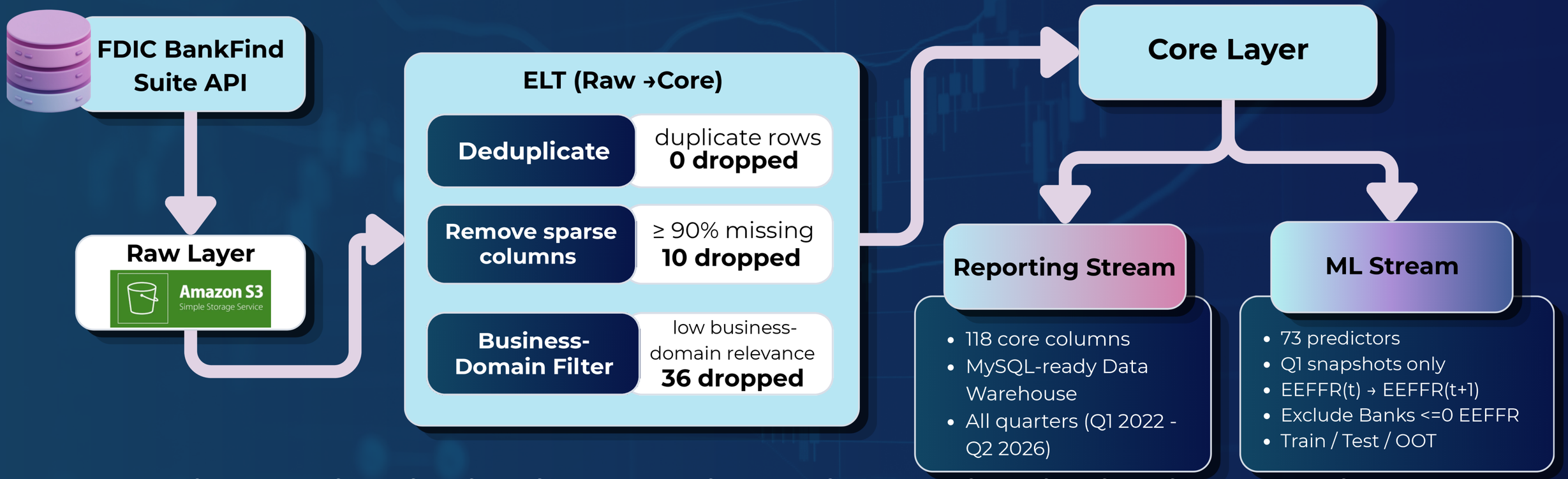
Distribution of EEFFR

Right-skewed

Data Challenges

- 88% rows contain at least one attribute beyond 3x IQR
- 20 columns >50% missing

Data Processing Pipeline & Governance



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Data Security Privacy & Audit Governance

Cloud Security: Private S3 bucket with Block Public Access active and SSE-S3 encryption enabled by default.

Least-Privilege IAM: Pipeline user scoped strictly to fdic; team assigned read-only roles.

Secrets Management: Credentials are kept in .env or an AWS CLI profile.

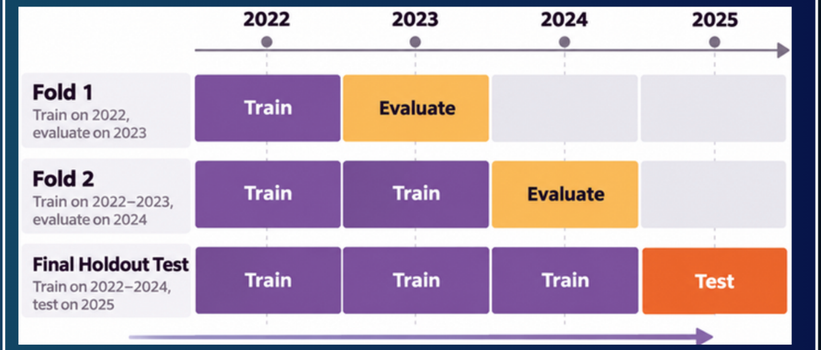
Audit Trail: Automated tracking via _manifest.json, core_column_decisions.csv, and _exclusions.csv.

Modelling Framework

Shared time-aware validation, target and data but different feature strategies

SHARED APPROACHES

Expanding-Window Cross-Validation



Applied to model comparison, hyperparameter tuning and feature-subset selection, depending on the model.

Data and Target

- **FDIC BankFind curated TEST and TRAIN with 73 candidate features**
- Prediction target (next-year Q1 EEFFR)
- Current-year EEFFR excluded in features

MODEL-SPECIFIC APPROACHES

Model		Feature Engineering	Feature Selection	Feature Reduction	Model Tuning
	Multiple Linear Regression	Financial ratios, log transforms + specialization dummies	Domain-based predictors + VIF diagnostics	Lasso evaluated; no final OLS reduction	OLS, Ridge, Lasso, log-log, Huber comparison
	Decision Tree	Numeric conversion, median imputation + log1p target	Permutation importance	Top-N feature subsets	Tree depth & leaf-size tuning
	Random Forest	3 engineered financial ratios	Mutual Information (MI)	73 → 25 features	Tree count, leaf size & feature sampling
	XGBoost	14 engineered ratios + log assets	Correlation ($r \geq 0.95$), leakage filtering + XGBoost importance	MAE-guided importance threshold (0.001) → 47 features	Boosting hyperparameters & pseudo-Huber loss

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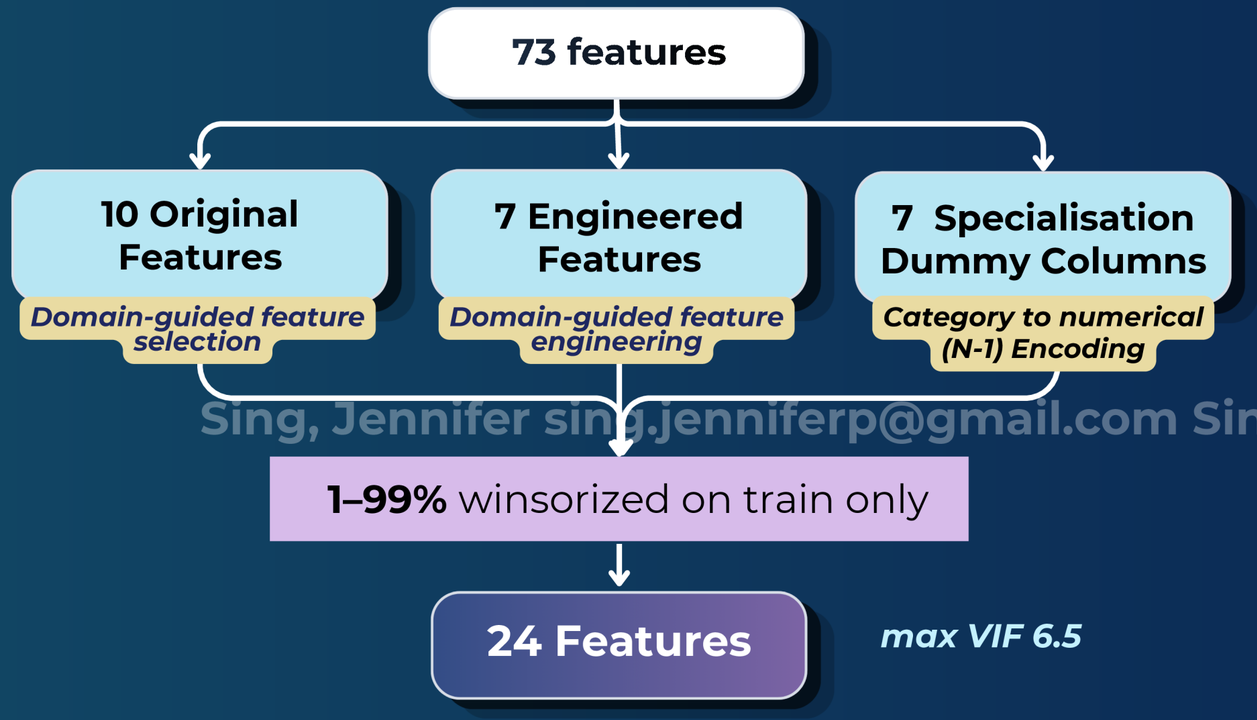
Multiple Linear Regression

Feature Processing, Model Selection and Accuracy

FEATURE PROCESSING

Log-transformed Target ER



ACCURACY ON THE TEST SET (Q1 2025 → Q1 2026)

Population	Banks	MAE	RMSE	R^2
All banks (primary)	4,287	12.304	38.915	0.101
Actual ER (0, 100%]	4,142	10.227	13.051	0.197
Actual ER (0, 200%]	4,270	10.998	14.943	0.287

Subgroups use actual outcomes and are retrospective diagnostics. All banks remain in the primary evaluation.

MODEL SELECTION & STABILITY

- **Model Tuning:** OLS, Ridge, Lasso, log-log, Huber comparison
- Same folds used as a stability check across years
- Fold MAE 12.24 (2023) · 14.58 (2024) · test 12.30: stable across years
- **Final: OLS on log(ER), HC3 robust standard errors**

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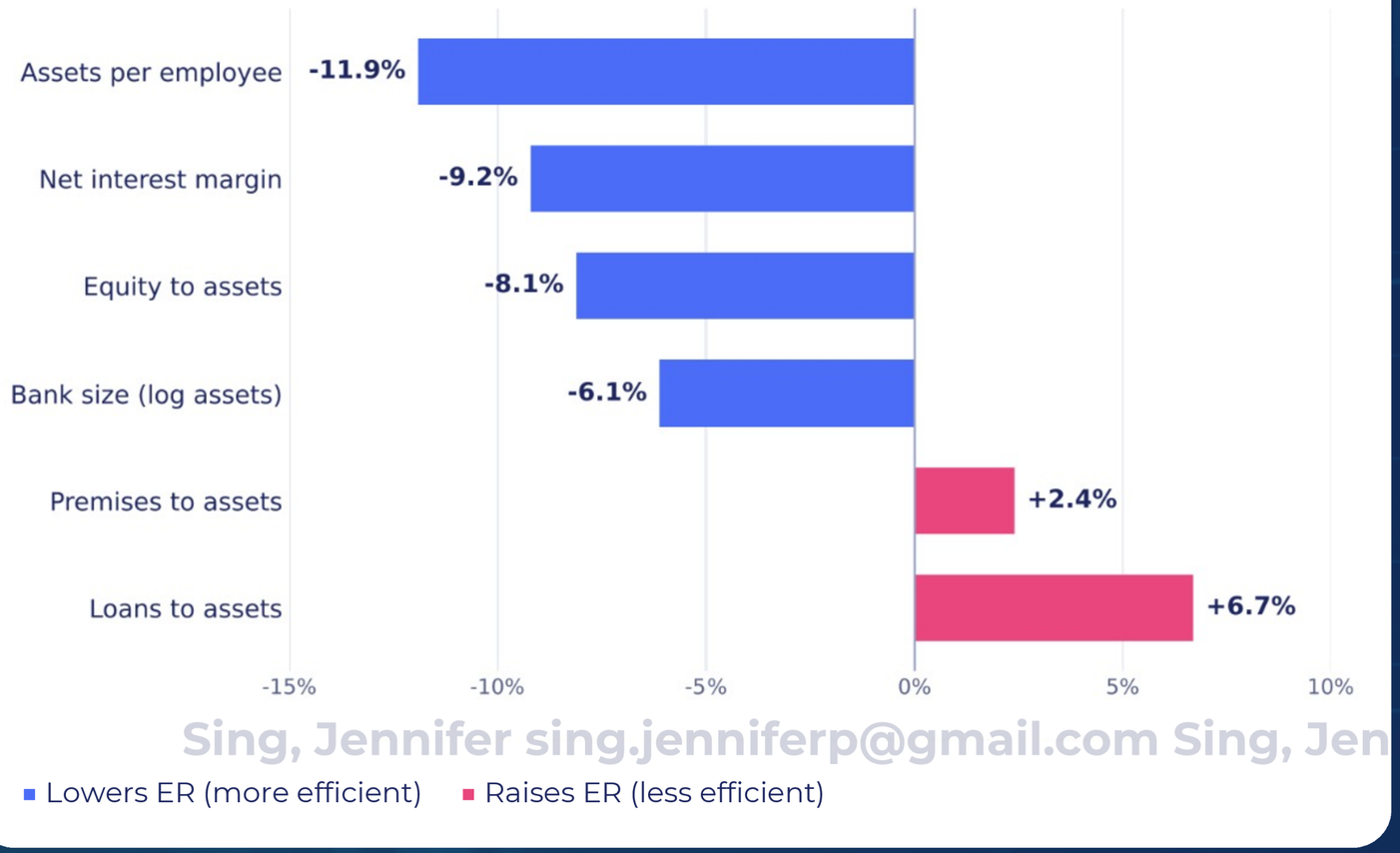
Multiple Linear Regression

What Drives Next Year's ER: Direction and Size



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Coefficients show association within the sample, not causal savings.

1 **Only MLR shows direction and size**
Log target, so effects read as % change in next-year ER. HC3 robust standard errors.

2 **Confirmed by tree models**
Net interest margin is top 3 in Decision Tree and XGBoost; staff productivity is top 3 in Decision Tree and Random Forest.

3 **Levers management controls**
Staff productivity, margin (pricing, deposit mix) and branch costs. Associations, not proven causes.

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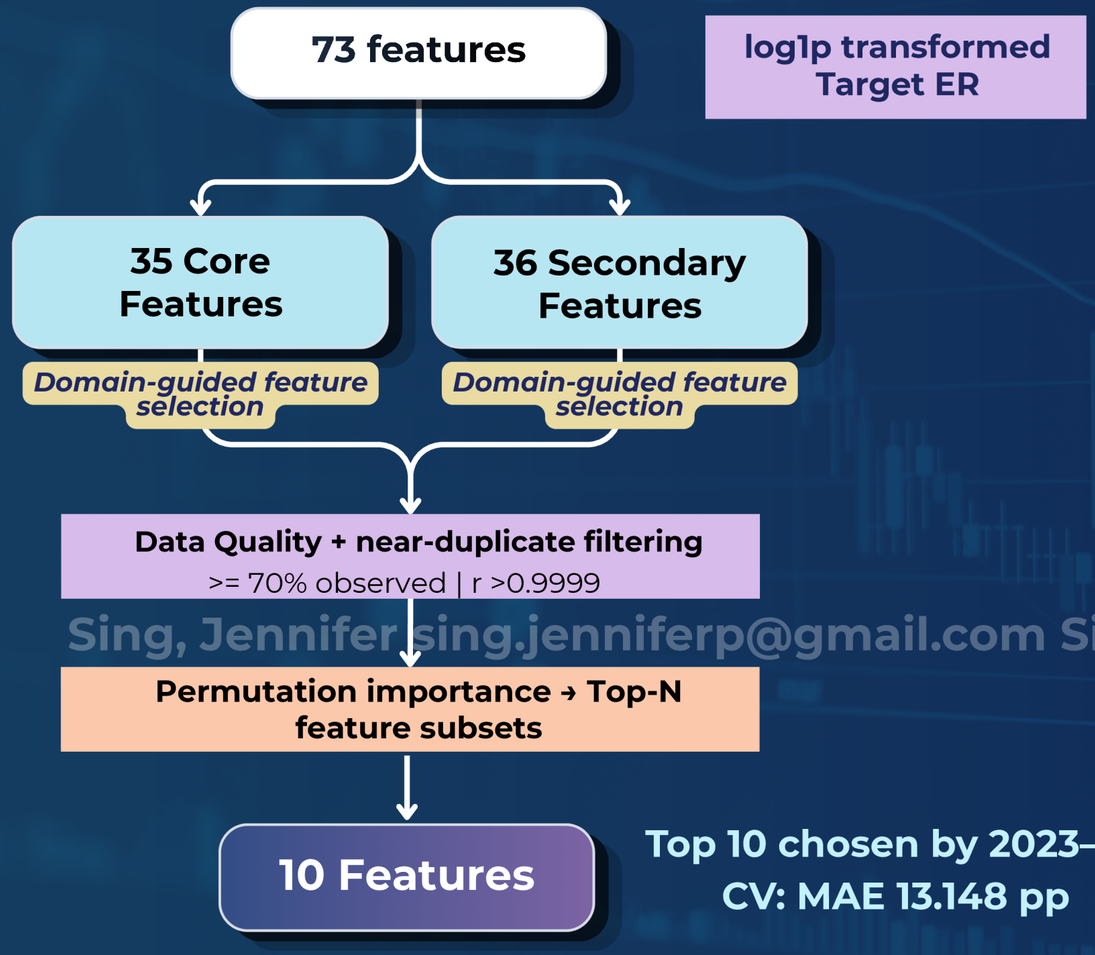
Decision Tree

FEATURE PROCESSING



ACCURACY ON THE TEST SET (Q1 2025 → Q1 2026)

Population	Banks	MAE	RMSE	R^2
All banks (primary)	4,287	11.350	38.808	0.106
Actual ER (0, 100%]	4,142	9.142	12.050	0.315
Actual ER (0, 200%]	4,270	10.005	14.215	0.355

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Decision Tree

FINAL DECISION TREE MODEL

10	100	10
Max Depth	Min. samples/leaf	Features

HYPERPARAMETER	FINAL Value	PURPOSE
min_samples_split	2	Min samples to consider a split
min_samples_leaf	100	Min samples required in each leaf
max_depth	10	Maximum depth of the decision tree

Objective: squared_error | scikit-learn 1.9.1

Target transformation: log1p

TOP 3 DRIVERS OF NEXT YEAR'S ER

1. **Assets per Employee** +3.768 pp MAE — Largest Model Reliance
2. **Net Interest Margin** +1.741 pp MAE — 2nd largest Model Reliance
3. **Income Taxes** +0.949 pp MAE — Additional predictive signal

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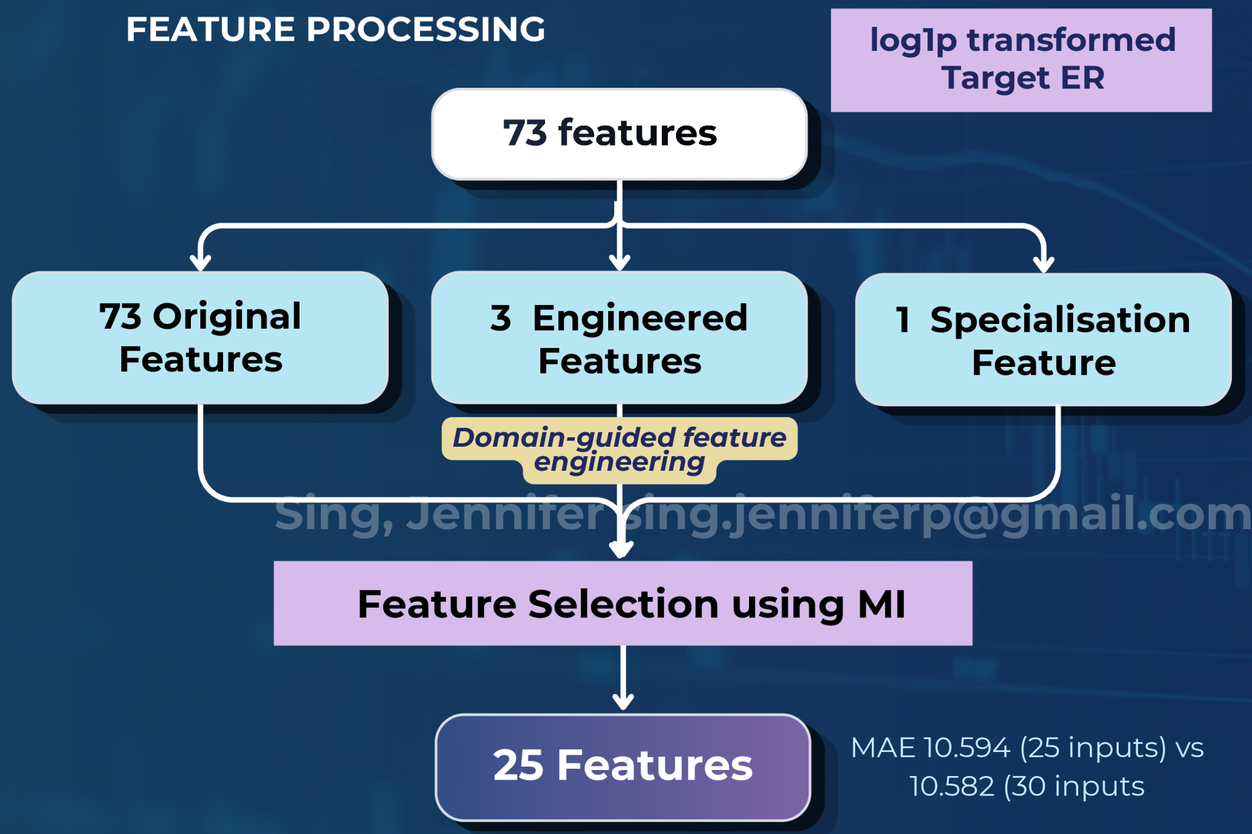
Random Forest

Feature Processing, Metrics and Final Model

FEATURE PROCESSING



200	5	25
Estimators	Min. samples/leaf	Features

HYPERPARAMETER	FINAL Value	PURPOSE
min_samples_split	2	Min samples to consider a split
bootstrap	true	Train trees on bootstrap samples
max_features	0.5	50% of transformed features considered at each split

Objective: squared_error | scikit-learn 1.9.1

Target transformation: log1p

ACCURACY ON THE TEST SET (Q1 2025 → Q1 2026)

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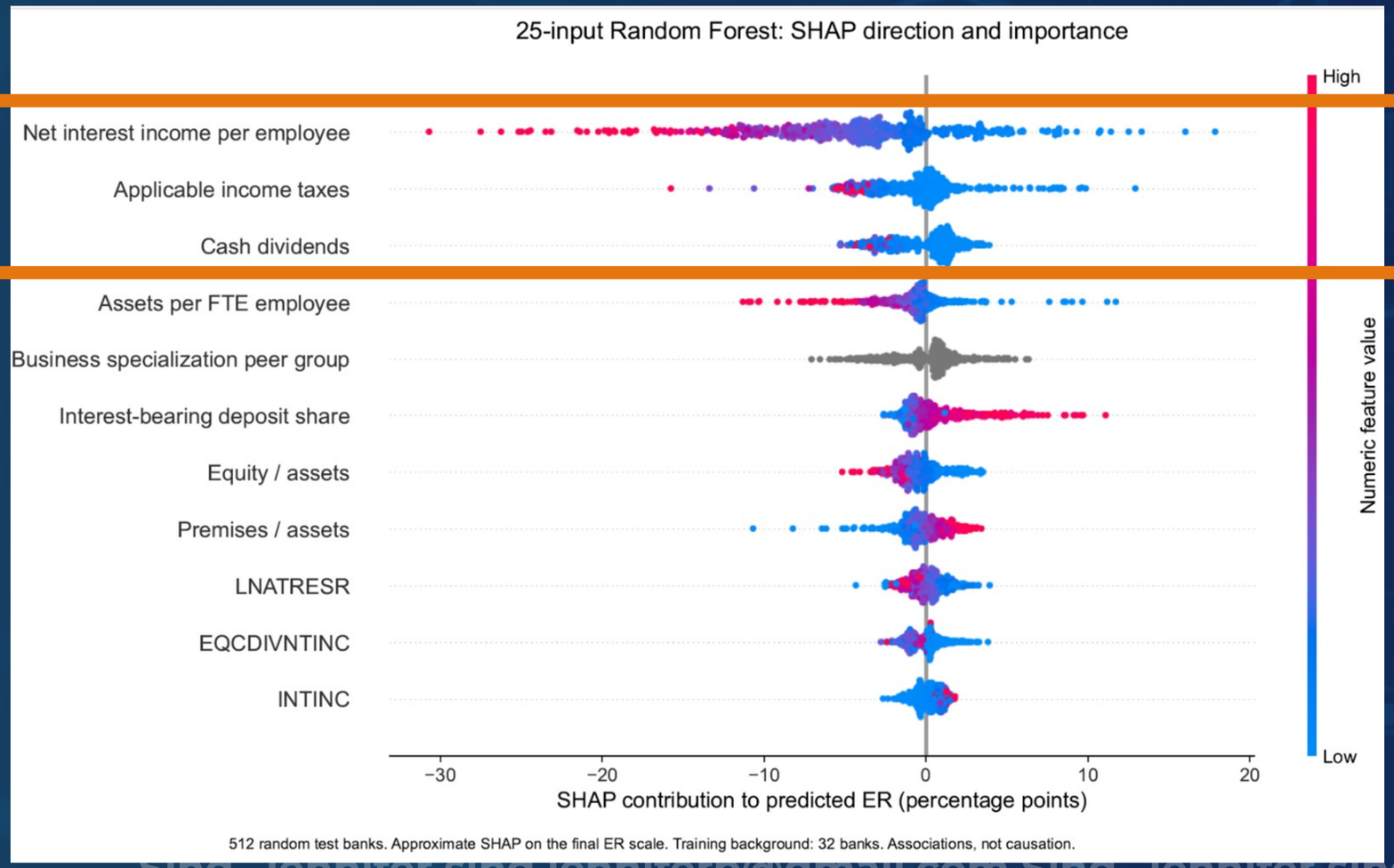
Population	Banks	MAE	RMSE	R^2
All banks (primary)	4,287	9.415	37.781	0.152
Actual ER (0, 100%]	4,142	7.366	9.833	0.544
Actual ER (0, 200%]	4,270	8.110	11.988	0.541

Peer-median benchmark MAE 14.076
Random Forest reduces MAE by 33.1%.

Random Forest

Top 3 Predictors of Next Year's ER (SHAP Analysis)



1 Net Interest income per employee

6.188 ER points

Higher employee productivity is associated with lower predicted ER.

2 Applicable income taxes

2.046 ER points

Higher taxes may reflect stronger profitability.

3 Cash Dividends

1.742 ER points

Larger dividend payments may signal stronger financial performance.

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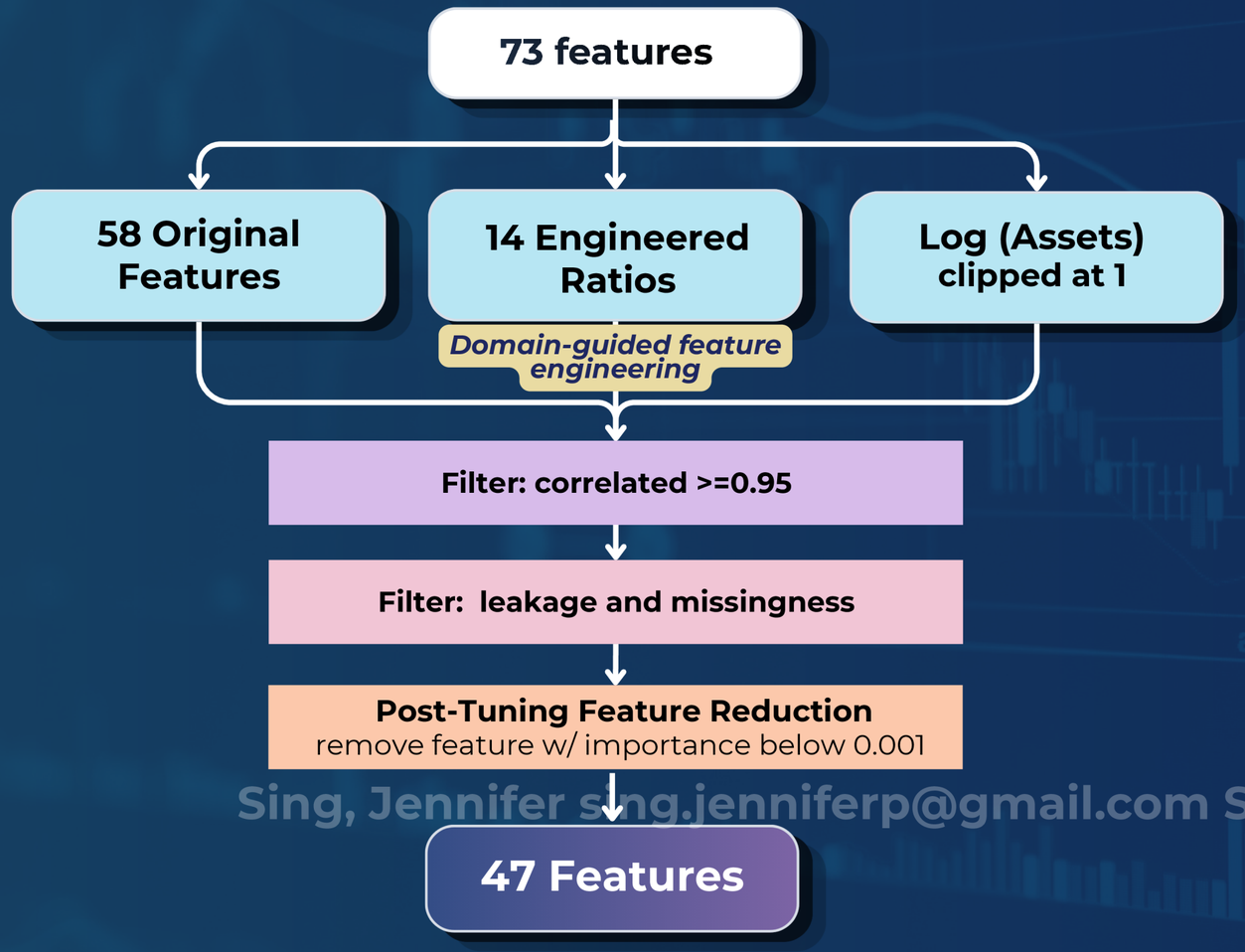
SHAP values do not imply a fixed change in ER for a one-unit change in the feature. The effect varies across observations.

XGBoost

Feature Processing, Metrics and Final Model

FEATURE PROCESSING



500	6	47
Estimators	Max depth	Features

HYPERPARAMETER	TUNED VALUE	PURPOSE
learning_rate	0.03	Gradual learning to limit overfitting
subsample	0.8	Uses 80% of training rows per tree
colsample_bytree	1.0	All features eligible for each tree
reg_alpha	0.1	L1 regularization
huber_slope	31.89	Controls robust loss sensitivity

Objective: reg:pseudohubererror | XGBoost library version: 3.4.1

ACCURACY ON THE TEST SET (Q1 2025 → Q1 2026)

Population	Banks	MAE	RMSE	R^2
All banks (primary)	4,287	9.29	37.204	0.178
Actual ER (0, 100%]	4,142	**7.43**	**10.988**	0.431
Actual ER (0, 200%]	4,270	8.088	12.654	0.489

MAE is lower for the retrospective ER ≤100 subgroup

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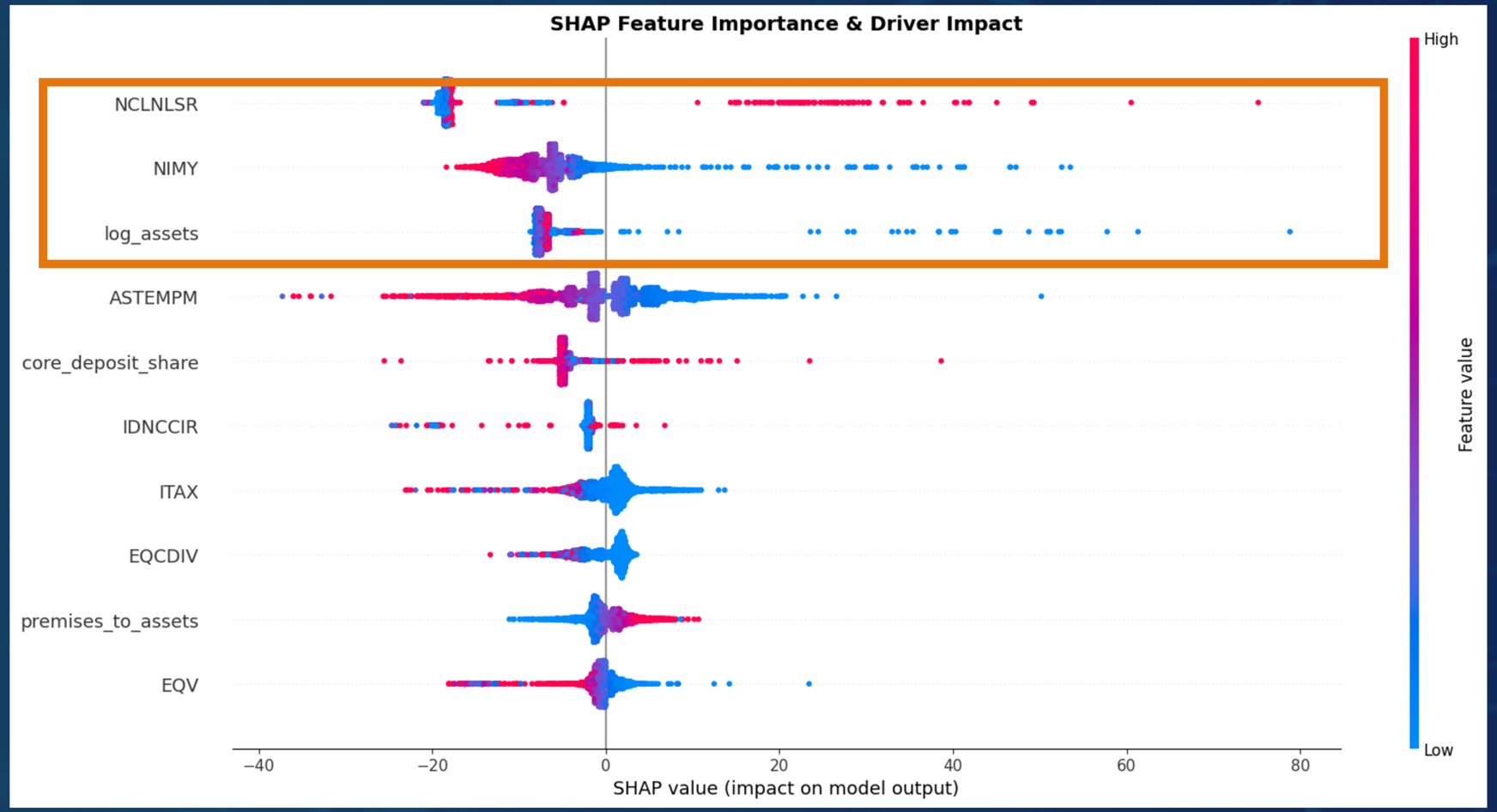
XGBoost

Top 3 Predictors of Next Year's ER (SHAP Analysis)



1 **Noncurrent Loans and Leases to Total Loans and Leases Ratio (%)**
18.46 ER points

Higher levels of delinquent or nonperforming loans may increase monitoring and servicing costs.

2 **Net Interest Margin (%)**
7.38 ER points

Stronger interest spreads may generate more income from earning assets.

3 **Bank Size (Total Assets)**
7.34 ER points

Larger banks can spread their costs across more assets, making them more efficient.

SHAP values do not imply a fixed change in ER for a one-unit change in the feature. The effect varies across observations.

XGBoost Leads Trained Models, but the Naïve Benchmark Remains Stronger

ACCURACY ON THE TEST SET (Q1 2025 → Q1 2026)

Model	MAE	RMSE	R^2 all	R^2 (actual ≤200)
XGBoost	9.29	37.20	0.178	0.489
Random Forest	9.42	37.78	0.152	0.541
Decision Tree	11.35	38.81	0.106	0.355
MLR	12.30	38.92	0.101	0.287
Naive benchmark	7.45	18.07	0.806	0.531

SUBGROUP PERFORMANCE BY PEER GROUP (MAE)

Peer group	Banks	XGBoost	RF	Naive
Commercial lending	2,352	8.00	8.54	7.08
Agricultural	922	8.17	7.71	6.55
Mortgage lending	309	12.19	11.78	9.85
Other specialized < $1B	**202**	**24.22**	**24.42**	**11.27**

Specialized banks show the highest prediction errors → prioritize manual review

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XGBoost Error Analysis: Three Key Model Limitations

1 Extreme Banks

17 banks caused 88% of squared error

All under-predicted; unusual or winding-down banks

actual next-year ER > 200 (17 banks)

2 Big Movers among Banks

44% of absolute error

MAE 21.5 vs 6.4 for stable banks

|next − current | > 10 (814 banks, 19%)
*includes deteriorating banks

3 Deteriorating Banks

underestimated deterioration by 41 points

Model is too optimistic about banks getting worse

ER got worse by more than 10 pts: next − current > 10

Why not use Naive to Predict?

0 of 225
deteriorating banks caught by naive

45 of 225
caught by XGBoost (+10 pt cut-off)

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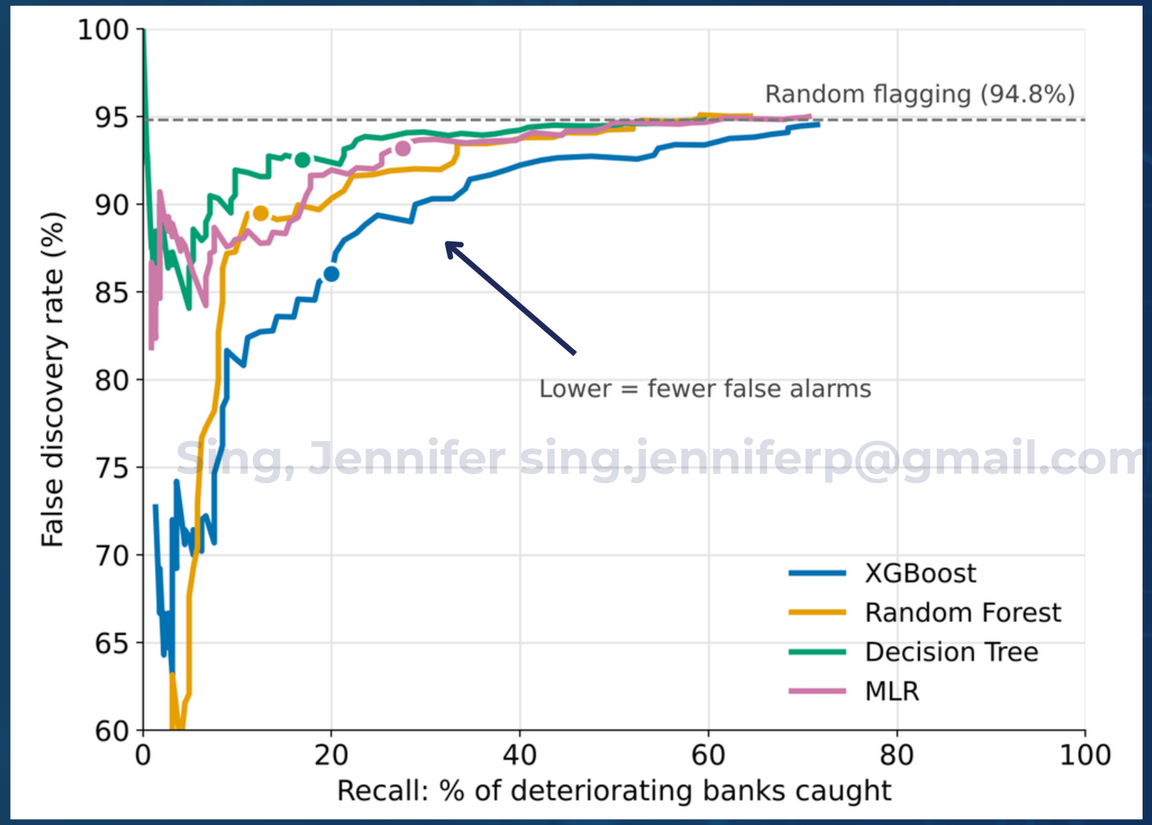
Naive assumes no bank changes, so it can never warn about a bank that gets worse. Supervisors need early warning, which we evaluate next.

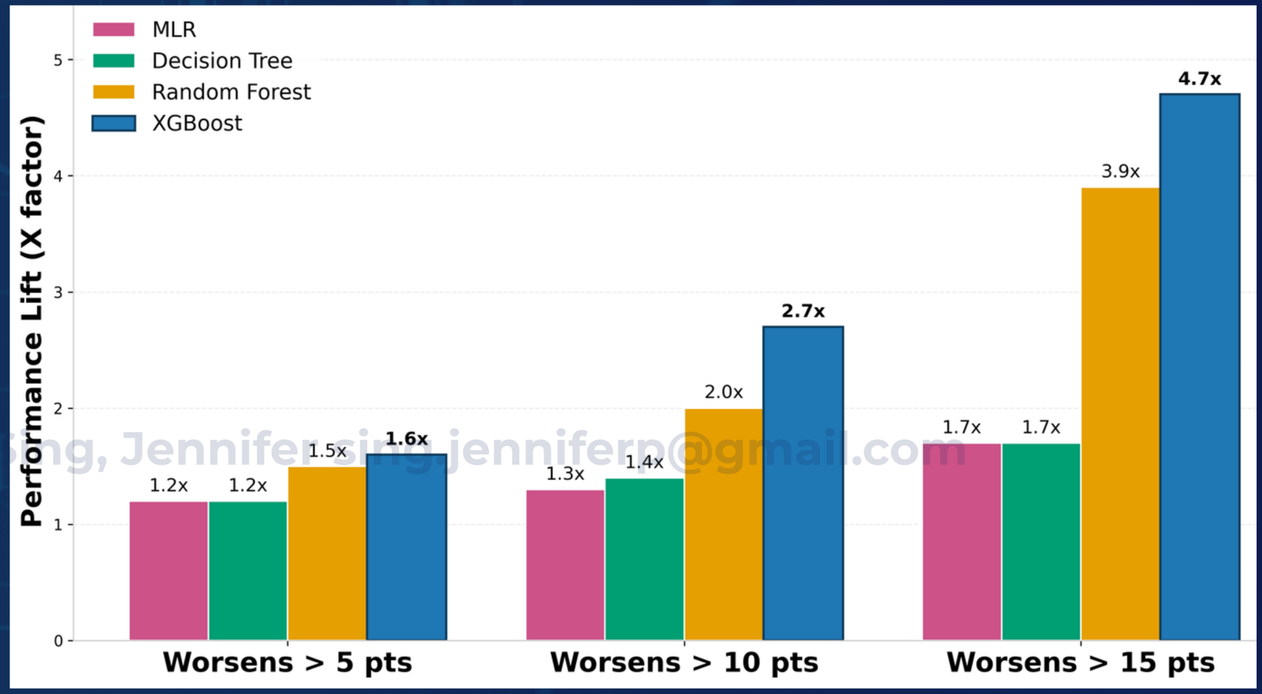
Early Warning Model: XGBoost Delivers Superior Lift

XGBoost catches more deteriorating banks with fewer false alarms

Recall vs False Discovery Rates Across Deteriorating Banks



XGBoost Ranks #1 Across All Severity Thresholds



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Deploying XGBoost: Balancing Early-Warning Recall and Review Capacity

Balancing supervisor review capacity with risk capture

Choosing the Decision Cut-off

Flag if rise >	Flagged	Caught	Recall	Hit rate
+10 pts	322	45	20%	14%
+5 pts	844	77	34%	9%
0 pts	1,803	123	55%	7%

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+10 pts (Default): Manageable workload (322 reviews), catches 20%, 2.7× better than random.

+5 pts (Higher recall): Catches 34% if review capacity allows (844 reviews).

0 pts (Aggressive): High coverage (55% recall), but overloads supervisors with 1,803 reviews.

Recap: XGBoost Model for Deployment

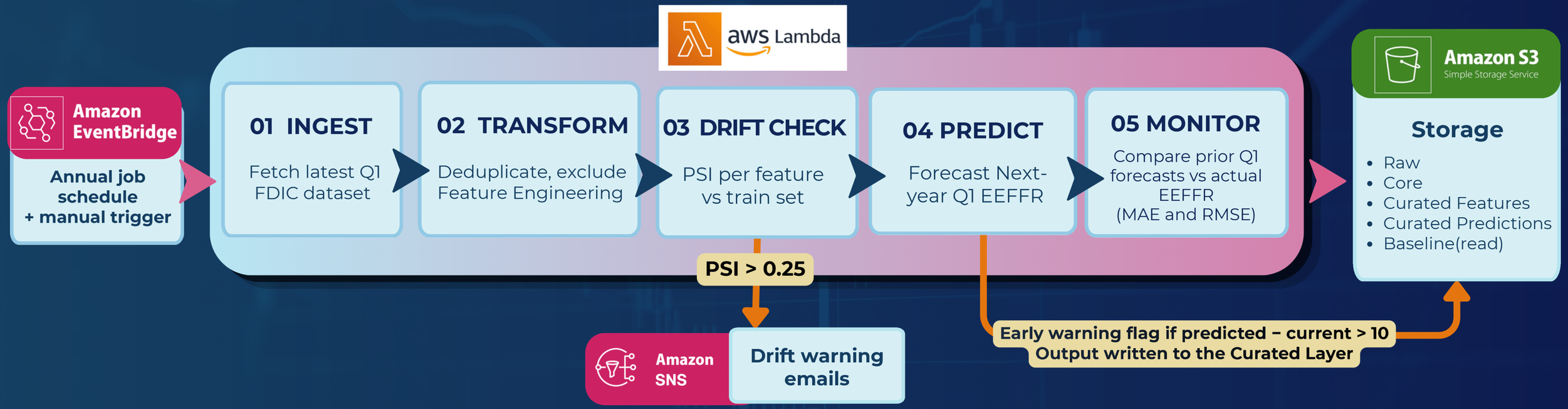
500	6	47
Estimators	Max depth	Features

HYPERPARAMETER	TUNED VALUE	PURPOSE
learning_rate	**0.03**	Gradual learning to limit overfitting
subsample	**0.8**	Uses 80% of training rows per tree
colsample_bytree	**1.0**	All features eligible for each tree
reg_alpha	**0.1**	L1 regularization
huber_slope	**31.89**	Controls robust loss sensitivity

Objective: reg:pseudohubererror | XGBoost library version: 3.4.1

AWS Batch Inference Architecture

Proactive identification of banks at risk of worsening operating efficiency



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01 WHY BATCH?

- Model uses annual Q1 snapshots.
- One run scores all eligible banks.
- Real-time inference adds no value.

02 EARLY WARNING

Flag predicted EEFFR deterioration of more than 10 points versus the bank's current ratio.

03 GOVERNANCE

- PSI > 0.25 triggers SNS warning.
- Monitor MAE / RMSE against the test baseline when actuals arrive.

Reconciling Drivers Across Models: From Linear Baselines to Non-Linear Realities

1. Net Interest Margin — 4 / 4 Models
2. Bank Size / Scale — 3 / 4 Models
3. Staff Productivity/ Assets per Employee — 3 / 4 Models
4. Tax Efficiency / Profitability — 3 / 4 Models
5. Capital Distributions / Dividends — 2 / 4 Models
6. Physical Overhead Intensity — 2 / 4 Models
7. Credit Quality / Nonperforming Loans — XGBOOST

Executive Leadership (Bank Management)

Targeted Productivity, Margin & Scale Levers

Investigate staff productivity and cost structure

Regulators & Bank Supervisors

Annual Predictive Screening & Exception Routing

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Credit & Risk Officers:

Early Credit Quality Tracking to Contain Servicing Costs

Monitor noncurrent loans closely

Investors & Acquirers

Peer Efficiency Gap Due Diligence

We restricted our direct recommendations to operational levers, such as staff productivity, overhead, and credit quality, where bank management and supervisors can take concrete, targeted action

Why Model Explainability is Vital in Banking

Stakeholder-Specific Interpretations

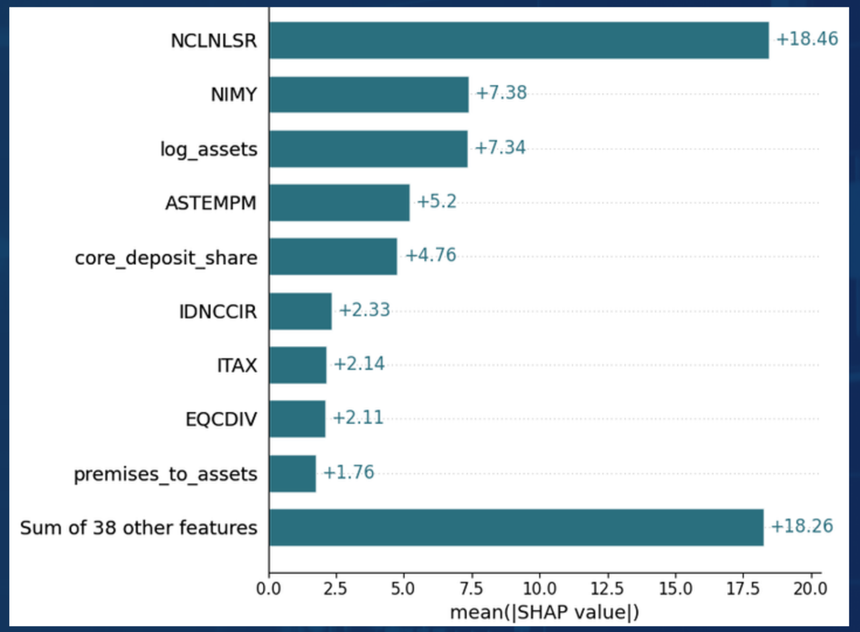
Regulators & Risk Officers
Identify key factors behind ER forecasts

Portfolio & Credit Managers
Annual risk screening across all institutions

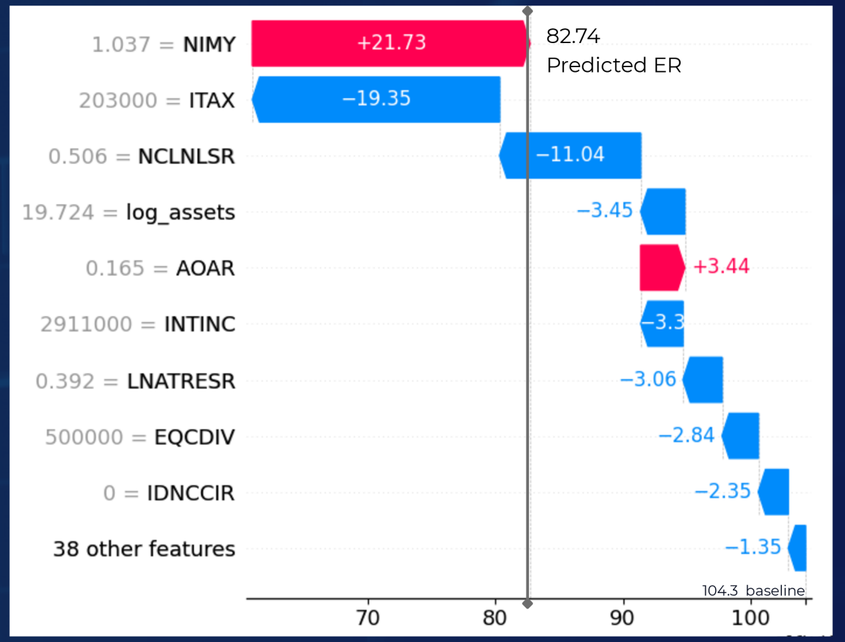
Executive Leadership
Guide targeted operational improvements

Audit & Governance Team
Trace feature contributions to individual predictions

XGBoost Global SHAP Importance



Individual Bank Waterfall Analysis



STATE STREET BANK&TRUST CO
Example

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3-Step Diagnostic Protocol

Global Feature Importance → **Individual Bank Explanation** → **Targeted Financial Review**

SHAP describes model contributions, not causal effects

Beyond Model Performance: Governance & Limitations

Intended Use

Data & Assumptions

Reproducibility

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Model Limitations

Early-Warning Trade-offs

Monitoring & Future Work

Thank you

Use of AI Tools

These slides incorporated the use of ChatGPT and Canva AI for generating illustrative images and Google Gemini for summarising ideas, finding options for slide titles, checking cohesion across slides, and spotting incorrect grammar.

All AI-generated outputs were critically reviewed, adapted, and edited to ensure accuracy, relevance, and academic integrity. The design, ideas, and overall direction remain own work